

The Pixels of Protection: Revealing Hidden Geometry of Corals Using Deep Learning and Photogrammetry to Support Reef Restoration

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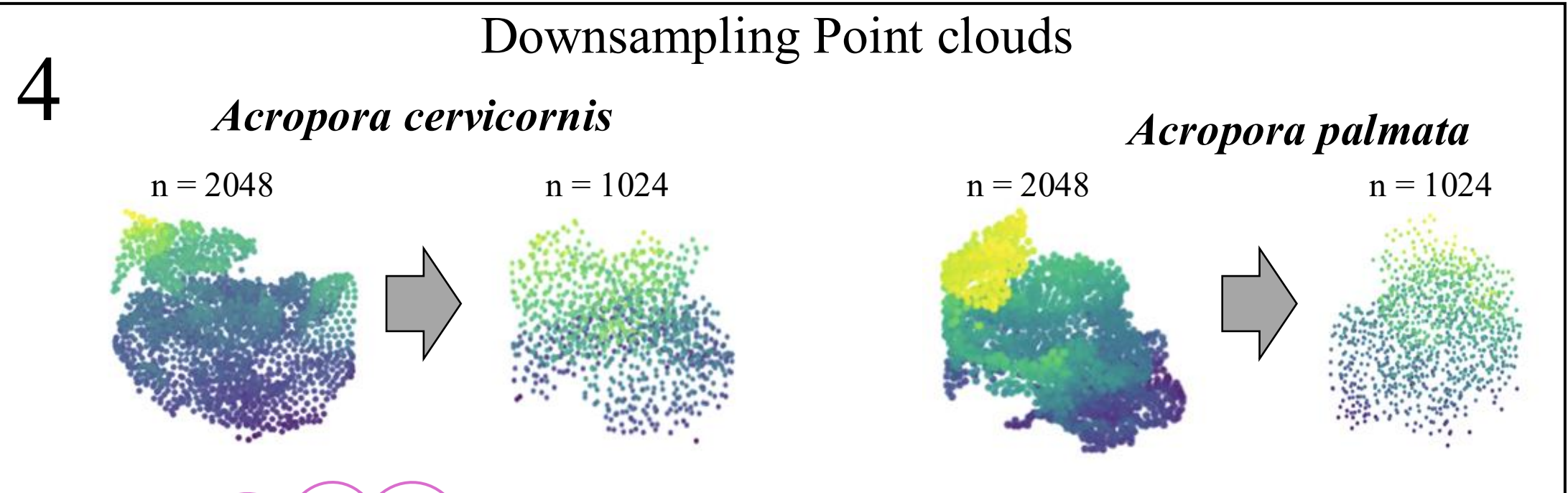
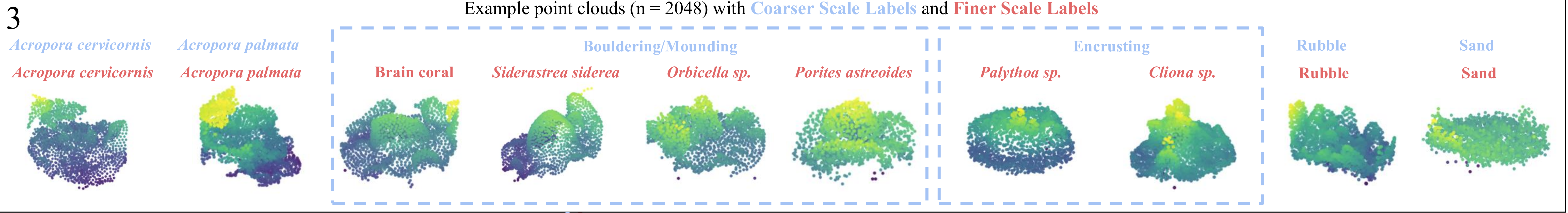
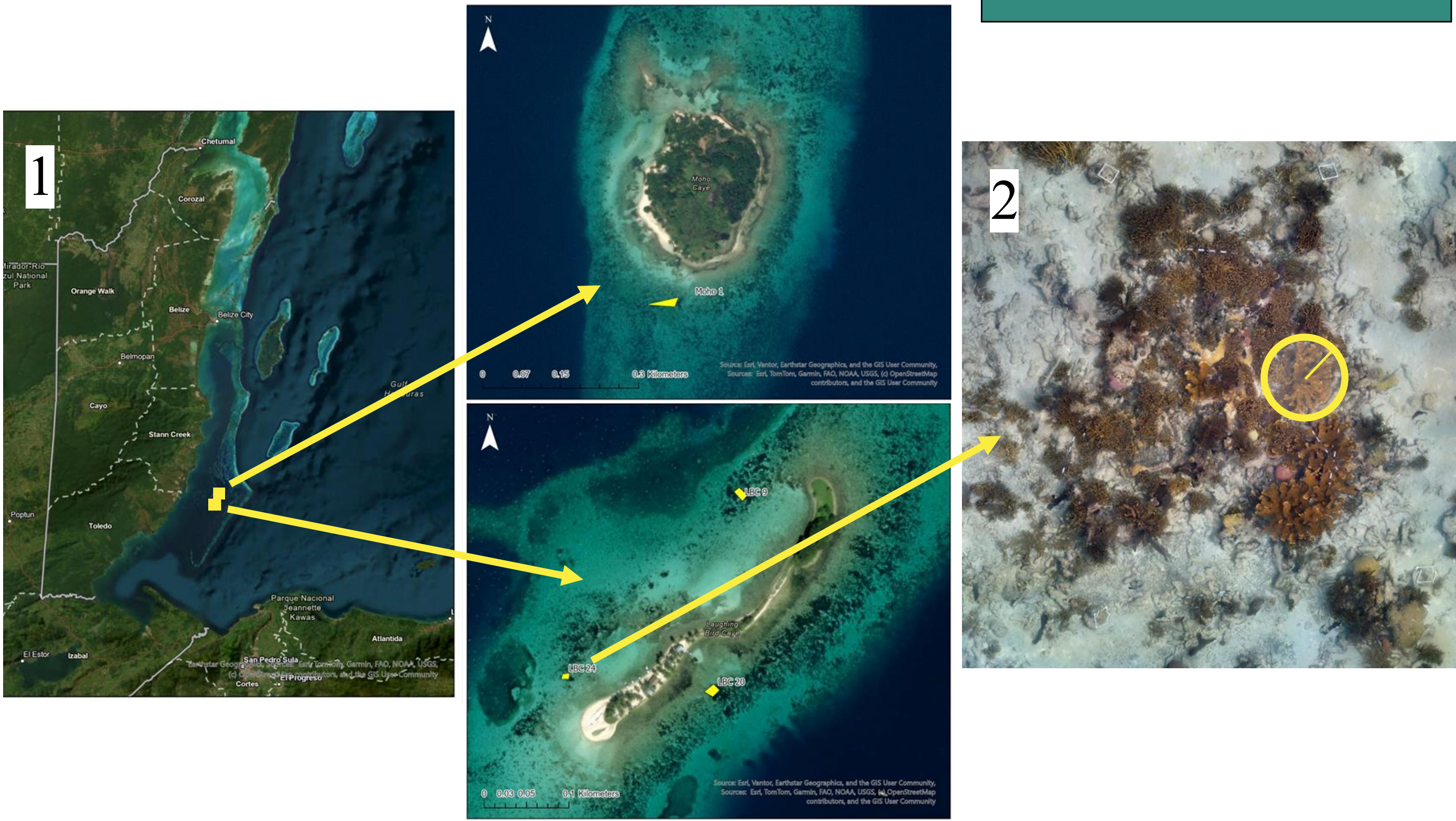


Motivation/Background

The ecosystem services of a coral reef largely relate to the structures created by coral, the foundational species, and other benthic components [1,2]. But how much do the geometric properties and complexity of these structures lead to the successful functioning of the reef ecosystem? And what do we stand to lose as corals decline on reefs? In this work, we investigate **how much information about coral restoration and biology can be learned from 3D point cloud geometry**. We looked at coral restoration data from southern Belize and evaluated the geometries of varying coral species and morphologies as well as rubble and sand geometries. As corals have declined, the geometries of rubble and sand are becoming more prevalent on reefs and therefore their structure compared to coral is more important to understand. Previously, coral reef complexity has been measured with chain-and-tape rugosity and calculations taken from landscape ecology like vector dispersion (rugosity) [3, 4]. However, we find that these metrics can fall short of capturing the changes in bottom structure clearly (visually) evident from photomosaic time series. Rather than weigh strengths and weaknesses of existing geometric calculations, here we have constructed an experiment to use machine vision to derive geometric features that can differentiate benthic structures.

Approach

Specifically, we adapted the Convolutional Neural Network (CNN) PointNet++ [5,6] based on previous successful applications to forestry data [7]. Our version of the model produces global feature vectors for different coral species and benthic components. The reef data for this model comes from a six-year time series of photomosaic point clouds capturing restoration site changes in Belize. The restoration work was done by Fragments of Hope, a nonprofit organization restoring coral since 2013 [8]. From the point clouds (XYZ data only), cylindrical “biopsies” of specific categories of coral and other benthic structures were isolated (Frame 2) then downsampled to a uniform size as inputs for PointNet++ (Frame 3). The model was then trained to accurately predict the structure category of individual biopsies. To investigate exactly how much geometric information PointNet++ can capture from photomosaic data, we varied both the amount of information going into and coming out of the model by changing the categorization (label) groupings (Frame 3), the downsampling size (Frame 4), and the global feature vector size (Frame 5).



Methods

1. Photomosaics were created in Agisoft Metashape from overlapping images for several restoration sites (~100 m²) outplanted by Fragments of Hope [8]. These photomosaics were from Laughing Bird Caye and Moho Caye in Belize and were taken between 2014 to 2020.
2. For each photomosaic plot, biopsies were collected across the benthic cover categories shown in Frame 3.
3. A random subset of 100 biopsies per finer category were then randomly selected to balance training, with data augmentation used to fill out rarer cover types (*Cliona*, *Palythoa*, *P. astreoides*, *A. palmata*).
4. Biopsies were downsampled using farthest point sampling [7] to various sizes including 1024, 2048 and 4096 and were labeled in the coarse or fine groupings in preparation for PointNet ++.
5. Data were randomly partitioned into training, test, and validation sets, then fed into an adaption of PointNet MSG ++ [5 -7].
6. The model is balanced using training and validation point clouds, then produces a global feature vector and predicted label for the test data (out-of-sample).

Results

- **Figure 1** examines test-set classification accuracy across numerical experiments. **A** and **B** show classifier confusion matrices for the best model parameters for finer scale and coarser scale labels, respectively. **C** displays how classification accuracy changes when varying the point cloud sizes after downsampling. **D** displays the how accuracy changes as the feature vector size is changed for fixed point cloud size (n = 1024 for coarser labels, n = 2048 for finer labels).
- **Table 1** shows the parameters for the two sets of label grouping that produced the overall highest classification accuracies.
- We find that for finer scales, more points and higher feature vector sizes are needed to improve accuracy, whereas coarser groupings resulted in higher accuracy with fewer points and with a smaller global feature vector.

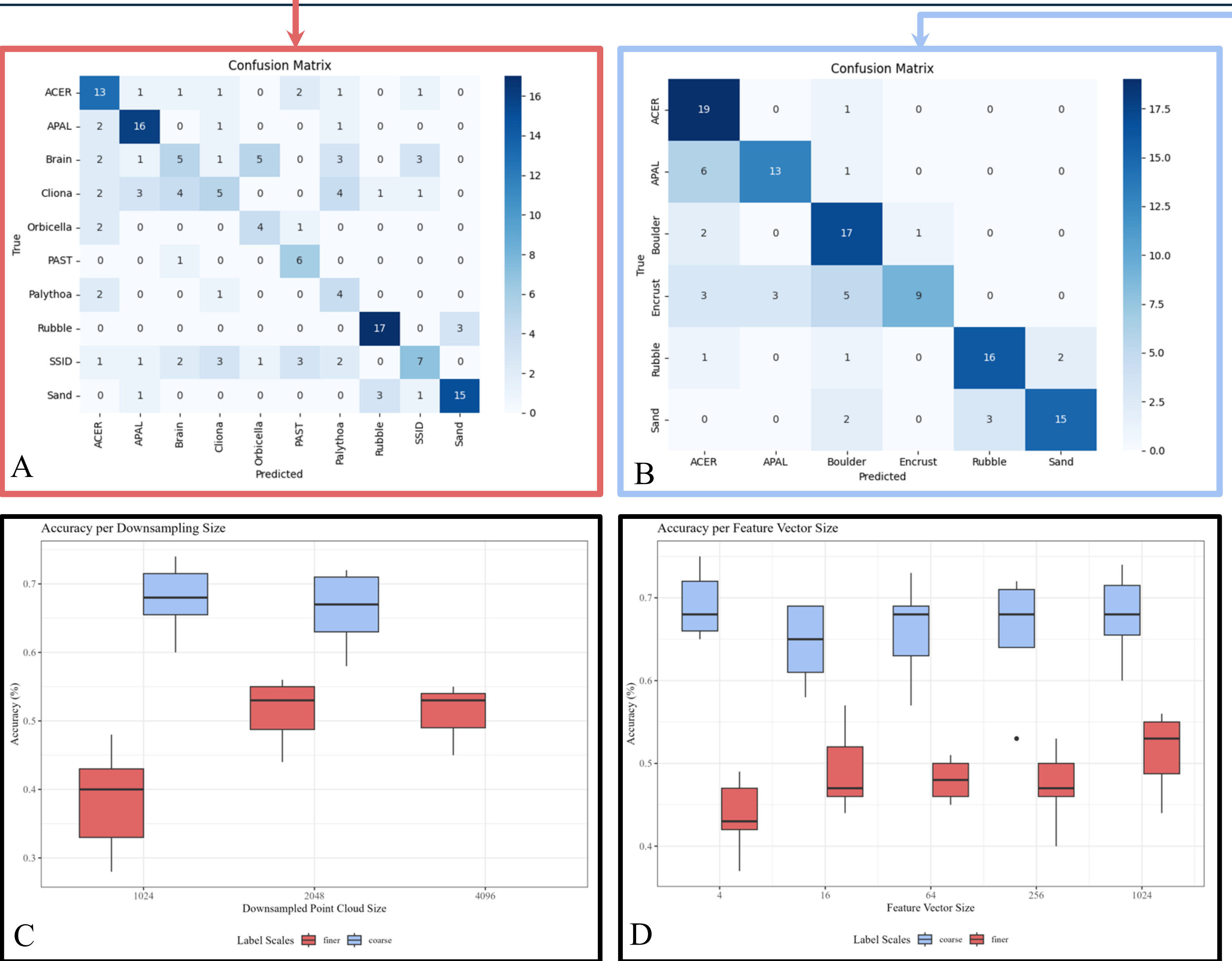


Table 1

Label grouping	Down-sampling size	Global Feature Vector Size	Best accuracy
Fine scale labels	2048	16	57%
Coarse labels	1024	4	75%

Remarks

This work sets the stage for formalizing new metrics that quantify the geometric information in 3D photomosaics to provide greater insight to coral restoration and conservation. Encouragingly, these results suggest that with sufficient density (n=2048) the biopsied point clouds contain sufficient fine-scale information to differentiate cover categories like *Cliona* and *Palythoa* that have similar gross geometric features (encrusting) but different texture. An important future direction is to test if the PointNet models trained thusly to categorize among cover types are learning geometric features that can distinguish meaningful variations within a coral group, like *A. palmata*, where small-scale geometric features can help determine genetic identity and possibly even predict differential environmental responses.

References



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